

Souradeep Banerjee

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F-1 Visa | Eligible for CPT & 36-month STEM OPT

EDUCATION

Arizona State University

Bachelor of Science, Computer Science (Concentration: Software Engineering)

- GPA: 3.5; Dean's List; New American University Scholar
- Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Operating Systems, Competitive Programming.

Tempe, Arizona

Expected May 2027

EXPERIENCE

Undergraduate Student Researcher

STAM Center at ASU

May 2026 – Present

Tempe, AZ

- Engineered a custom **LLVM optimization pass** for dynamic Intermediate Representation (IR) instrumentation, profiling runtime execution frequencies across 6 distinct architectural instruction categories.
- Decreased instrumented binary size and execution overhead by optimizing state updates, aggregating per-instruction **load/add/store** routines into singular **basic-block terminator injections**, validating performance via the **EmBench-iot suite** across O0–O2 optimization levels.
- Investigated capability-aware hardware-software co-design—a security model replacing plain memory addresses with metadata-bound access permissions—by analyzing the **Zeno Compiler's** transparent, **Namespace-based (NSID)** translation for secure **RISC-V architectures**.

Software Engineering Intern

Cruw.org

Jan 2023 – Jan 2024

Kolkata, India (Remote)

- Delivered **2** full-scale client-side applications with a consistent **60 FPS** baseline and **30%** fewer UI bugs by architecting robust state management and optimizing custom animation loops in Flutter.
- Accelerated feature development timelines by **20%** by systematically evaluating, debugging, and integrating **5** complex open-source Dart libraries into proprietary codebases.

PROJECTS

Cyclops MIPS (Cycle-Accurate CPU Emulator) | C++23, ImGui, GTest, CMake | [Demo](#)

Nov 2025 – Present

- Engineered a cycle-accurate 5-stage MIPS32 CPU simulator in **C++23**, sustaining optimal **Instruction Per Cycle (IPC)** by mitigating pipeline flushes and data hazards via a **256-entry Branch Target Buffer (BTB)** and custom **EX/MEM forwarding paths**.
- Streamlined instruction execution and memory latency by developing a custom **two-pass assembler** using `std::expected` for zero-overhead error handling, integrated with **4KB direct-mapped L1 caches** featuring write-allocate policies.
- Validated hardware correctness and single-cycle execution precision by architecting a comprehensive **Google Test (GTest)** and **Google Benchmark suite**, ensuring deterministic behavior across 20+ instruction edge cases.

Cyclops Studio (High-Performance Animation Engine) | C++17, OpenGL 4.6, ImGui, CMake | [Demo](#)

Oct 2025 – Present

- Achieved stable **4K/144 FPS** rendering by architecting a custom **OpenGL batched quad renderer** (10,000 quads/call), preventing **VRAM exhaustion** via an **FBO-based zero-copy compositing pipeline** that quantizes GPU memory per keyframe.
- Minimized I/O latency and executed sub-millisecond flood-fills by implementing an $O(W \times H)$ **scanline BFS algorithm** for bulk texture uploads, coupled with a custom **miniz zlib-deflate binary serialization format** for RGBA8 compression.
- Eliminated main-thread UI blocking during video export by engineering a **multi-threaded render pipeline**, using `glReadPixels` to pipe frame buffers directly to background `ffmpeg` subprocesses.

Joint DiT (Multimodal Diffusion Model) | Python, PyTorch, CUDA

Jul 2025 – Oct 2025

- Engineered a multimodal diffusion architecture, expanding maximum sequence length by **300%** without Out-Of-Memory (OOM) crashes, by implementing aggressive **KV-cache optimization** and query chunking on single-GPU setups.
- Reduced cross-modal matrix computation overhead by **40%** via custom, memory-efficient **attention masks**.
- Accelerated iteration cycles by automating deployment and documentation, saving an estimated **5 hours per week** in developer overhead, by establishing a robust **CI/CD pipeline** and Gradio interface.

LEADERSHIP & ACHIEVEMENTS

1st Place Hackathon Wins: ASA DataFest (Best Insight) & VillageHacks

Architected algorithmic solutions and data pipelines to secure top inter-university placements.

Mar 2026 – Apr 2026

Tempe, AZ

ICPC Rocky Mountain Regional Finalist & NAQ 2nd Place Finisher

Secured 2nd Place locally (5 problems solved); advanced to represent ASU against top regional teams.

Oct 2025 – Nov 2025

Tempe, AZ

Manager of Special Programs, ACM at ASU

Instruct ICPC Primer workshop series on advanced algorithms, Segment Trees, and Disjoint Set Union.

Oct 2025 – Present

Tempe, AZ

Grand Challenges Scholar (Secure Systems) & CodePath Advanced SWE (Top 5%)

Research secure HW/SW architecture; earned Advanced certification mastering graph & DP algorithms.

Aug 2024 – Present

Tempe, AZ

TECHNICAL SKILLS

Languages: C++ (17/20/23), Rust, C, Assembly (MIPS, RISC-V), Python, SQL, Java, Dart, JavaScript/TypeScript

Systems & Cloud: Linux (Kernel/User), Docker, Kubernetes, AWS, CMake, GDB, CI/CD, Git

Libraries & Frameworks: LLVM, OpenGL, ImGui, STL, PyTorch, React.js, Node.js, Flutter

Concepts: Computer Architecture, OS Internals, Concurrency, Memory Management, Distributed Systems